

Yavuz Shahzad

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in yavuz-shahzad 📺 yshahzad

Skills

Programming Languages: Python, R, SQL, $\text{\LaTeX}$.

Libraries and Frameworks: Polars, Pandas, PySpark, Scikit-learn, Seaborn, Tidyverse.

Developer Tools and Platforms: GitHub, Databricks, Azure, Jupyter Notebook, Excel.

Non-technical: Outstanding communication skills; strong collaborative skills and attention to detail.

Education

McGill University

Montreal, QC

MSc in Biostatistics

2026 - 2028

- **Research area:** Causal machine learning for heterogeneous treatment effect estimation.

McGill University

Montreal, QC

BA in Statistics, Minor in Psychology

2022 - 2026

- **Coursework:** Mathematical Statistics I & II, Statistical Learning, Experimental Design, Sampling Theory, Generalized Linear Models, Honours Analysis I & II, Honours Algebra I & II.

Experience

Undergraduate Researcher

Montreal, QC

McGill Clinical and Health Informatics Group

Aug 2025 – Present

- Part of the Data-Driven Decision Modeling (D3Mod) Lab, developed supervised machine learning models (NGBoost) for probabilistic prediction of individualized hemoglobin recovery after blood donation.

Data Science Intern

Toronto, ON

Manulife

May 2025 – Aug 2025

- Developed time series models (Prophet) to forecast new regular premiums for life insurance products, reducing MAPE by 45% compared to legacy models.
- Conducted exploratory data analysis to recommend new lead types for proactive outreach to financial advisors and implemented the delivery of leads to wholesalers in a production PySpark pipeline.
- Presented leads pipeline and forecasting models to senior leadership, incorporating stakeholder feedback.

Undergraduate Researcher

Montreal, QC

Brain Research and Imaging of Neurorehabilitation Lab: McGill University

Jan 2023 – Sept 2024

- Cleaned and analyzed data, conducted literature review, and drafted a manuscript about a novel treatment paradigm for those with chronic stroke.
- Analyzed neural markers of long-COVID survivors using a threshold-based β -burst detection algorithm.
- Served as a consultant to colleagues, refining analysis methodologies to ensure statistical rigor.

Research Intern

Boston, MA

J. Philip Kistler Stroke Research Center: Harvard Medical School

Jun 2022 – Aug 2022

- Performed data abstraction from the electronic health records of over 200 patients. Study assessed the efficacy of left atrial appendage closure in preventing brain hemorrhages in atrial fibrillation patients.

Publications

- Milot, M.-H., Palimeris, S., **Shahzad, Y.**, Corriveau, H., Tremblay, F., Boudrias, M.-H.
Long-term benefits of a tailored strength training intervention on arm function in chronic stroke survivors: a follow-up study. **Journal of Rehabilitation Medicine – Clinical Communications**, Vol. 8, February 2025.
<https://doi.org/10.2340/jrm-cc.v8.42941>

Awards and Honours

- NSERC Undergraduate Student Research Award (\$8,800)** 2024, 2026
- Awarded in 2024 for research under Dr. Marie-Hélène Boudrias of McGill's BRAIN Lab.
 - Awarded in 2026 for research conducted under the supervision of Dr. Alton Russell, applying Neural Stochastic Differential Equations to predict ICU patients' risk of requiring hemoglobin transfusion.
- Dean's Honour List** Apr 2022

Community Involvement

- Editor: McGill Science Undergraduate Research Journal** 2022 - 2026
- Facilitate the peer review of manuscripts and ensure the journal's standards of scientific rigour are consistently upheld.
 - As former Managing Editor, lead MSURJ's outreach portfolio and ensured that the journal received numerous manuscript submissions.
- Council Member: SUMS McGill** 2024 - 2025
- Participated in and voted during the meetings of McGill University's Society of Undergraduate Mathematics and Statistics Students (SUMS).

Work Eligibility

Citizenships: USA, Canada, Turkey.